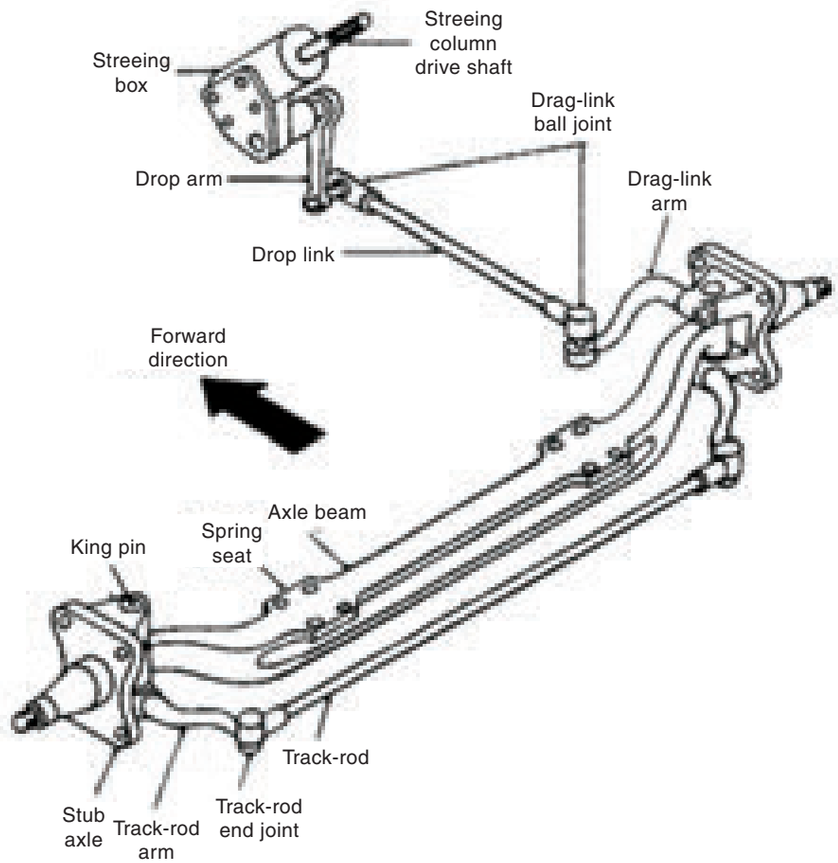


HNSVNC

Figure 8.2.1 (b) Steering system

8.3.1.1 Rack and pinion steering gear box

Construction

- In this type of steering gearbox, in place of cam or worm, the pinion gear is attached to the steering shaft through universal joints.
- The pinion is meshed with rack gear. The circular input motion of the pinion is converted into a linear rack output movement.
- The tie rod at each end of the rack is connected to the steering arm by means of ball joint.

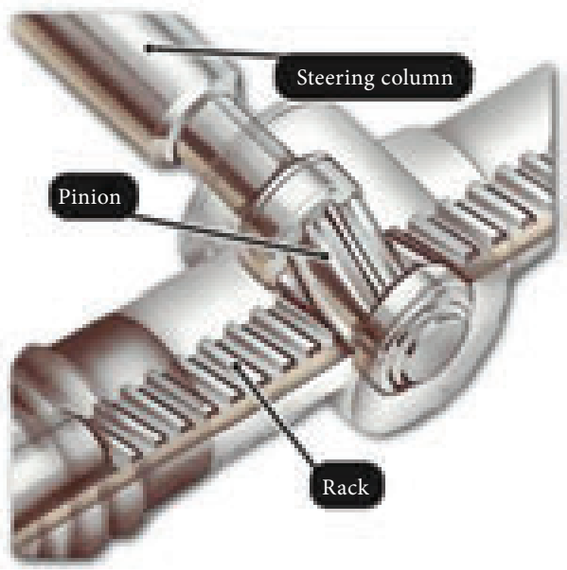


Figure 8.3.1.1 (a) Rack and pinion type steering gear box

Working Principle

- When the drive turns the steering wheel, the pinion at the end of steering wheels rotates at its own axis.
- Based on the direction of rotation of the pinion, the rack moves towards the left or right.

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